

Bool Tutorial Sheet

1. Complete the truth table for AND (2 input, 3 input, 4 input) and draw the logic symbol for each of them.
2. Complete the truth table for OR (2 input, 3 input, 4 input) and draw the logic symbol for each of them.
3. Complete the truth table for the Inverter and draw the logic symbol.
4. Complete the truth table for NAND (2 input, 3 input, 4 input) and draw the logic symbol for each of them.
5. Complete the truth table for NOR (2 input, 3 input, 4 input) and draw the logic symbol for each of them.
6. Complete the truth table for XOR gate and draw the logic symbol.
7. Complete the truth table for XNOR gate and draw the logic symbol.
8. Complete the Truth Table for the following exercises:
 - i) $Z = A + BC$
 - ii) $Z = AB + \bar{C}$
 - iii) $Z = ABD + A\bar{C}$
 - iv) $Z = \overline{A + C} + \overline{BC}$
 - v) $Z = \bar{A}C + (B \oplus C)$
 - vi) $Z = \overline{A \oplus C} + \overline{A + B}$